TRUE Simulation Experiment Report

P0/P1 Refactored Round

Trust-Reliability Unified Engineering
Multi-Agent Collaboration Simulation

100 runs x 200 rounds x 4 scenarios x 7 groups
Fixed task-flow pairing | Bootstrap CIs | Bonferroni correction

Executive Summary

This report presents the P0/P1 refactored experimental results for the TRUE (Trust-Reliability Unified Engineering) multi-agent collaboration framework. All groups within each Monte Carlo run face the identical task sequence, ensuring that outcome differences are attributable to selection mechanisms rather than task luck.

Key changes: (1) Fixed task-flow pairing; (2) Removed group-specific observation manipulation for A8; (3) Renamed MOO to TTB (Trust-Targeted Baseline) and added a proper MOO baseline using weighted Tchebycheff; (4) Added ablation variants TRUE-C, TRUE-E, TRUE-N; (5) Wilcoxon signed-rank test + bootstrap 95% CIs + Bonferroni correction.

Scenario Definitions

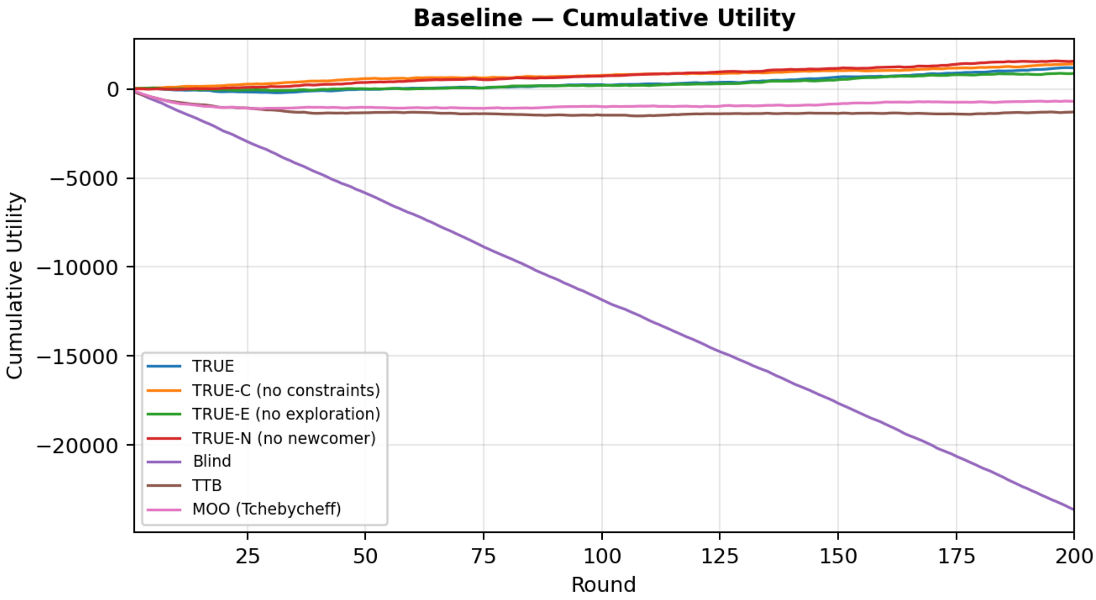
Scenario	Description
Baseline	Standard engineering collaboration with moderate observation noise.
Safety-Critical	High-stakes safety environment with harsher penalties and stronger error propagation.
Observation-Manipulated	Observation-contaminated environment where surface quality diverges from true quality.
Utility-Trust-Misalignment	Local phases where trusted incumbents underperform and low-history entities hold latent value.

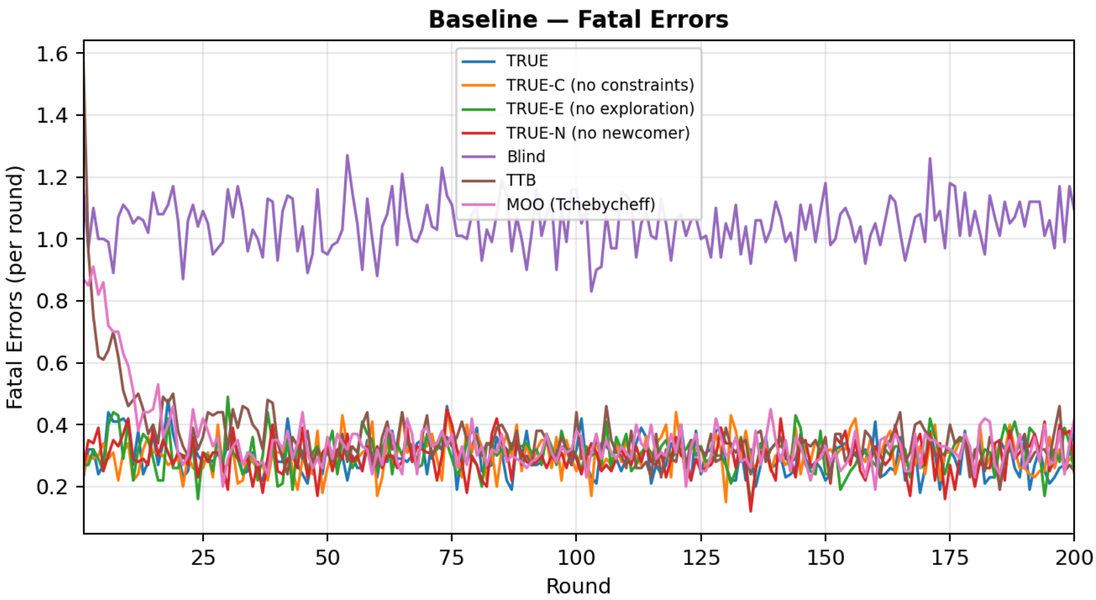
Results Summary

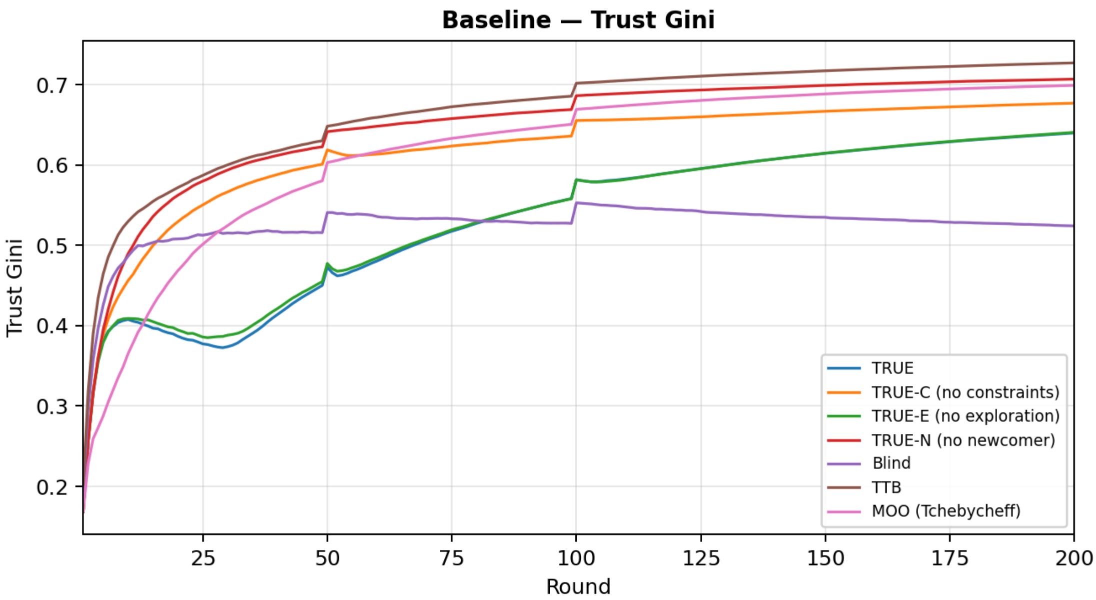
Mean cumulative utility, 95% bootstrap CI, fatal errors, and key metrics per scenario and group.

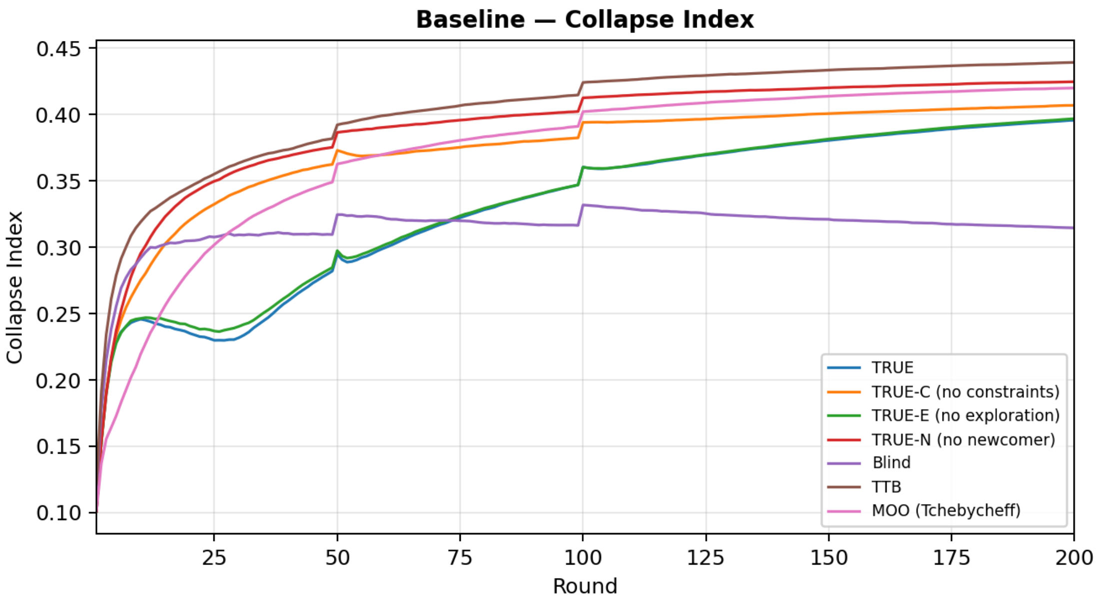
Scenario	Group	Cum.Utility	95% CI	Fatal	True Q	Gini	Collapse	A9 Delay
Baseline	TRUE	1185.0	[865.4, 1506.8]	59.3	0.769	0.640	0.396	0.2
Baseline	TRUE-C (no constraints)	1407.6	[1064.4, 1752.4]	60.4	0.800	0.677	0.407	0.8
Baseline	TRUE-E (no exploration)	872.2	[500.8, 1228.2]	61.2	0.767	0.641	0.397	0.2
Baseline	TRUE-N (no newcomer)	1570.0	[1209.4, 1912.0]	59.5	0.802	0.707	0.425	465.2
Baseline	Blind	-23644.8	[-23935.4, -23335.2]	209.6	0.342	0.524	0.315	7.0
Baseline	TTB	-1289.8	[-2340.8, -369.2]	72.0	0.741	0.727	0.439	700.4
Baseline	MOO (Tchebycheff)	-695.8	[-1442.4, -32.0]	69.5	0.767	0.699	0.420	790.3
Safety-Critical	TRUE	-3596.2	[-4179.6, -3014.4]	41.7	0.796	0.654	0.413	0.3
Safety-Critical	TRUE-C (no constraints)	-2034.6	[-2543.0, -1504.4]	39.4	0.828	0.688	0.420	0.9
Safety-Critical	TRUE-E (no exploration)	-3580.0	[-4218.2, -2915.6]	42.0	0.796	0.657	0.416	0.6
Safety-Critical	TRUE-N (no newcomer)	-1677.0	[-2187.2, -1166.4]	38.4	0.833	0.724	0.440	959.4
Safety-Critical	Blind	-53594.0	[-54118.8, -53063.4]	212.6	0.345	0.538	0.323	7.8
Safety-Critical	TTB	-5979.4	[-7357.8, -4786.8]	48.4	0.782	0.733	0.446	751.6
Safety-Critical	MOO (Tchebycheff)	-4972.8	[-5730.0, -4202.0]	46.0	0.803	0.721	0.438	840.0
Observation-Manipulated	TRUE	1815.6	[1488.4, 2130.8]	52.5	0.769	0.605	0.365	0.0
Observation-Manipulated	TRUE-C (no constraints)	2517.2	[2184.0, 2867.2]	51.2	0.803	0.658	0.395	0.8
Observation-Manipulated	TRUE-E (no exploration)	2270.8	[1973.2, 2560.4]	51.0	0.770	0.611	0.368	0.0
Observation-Manipulated	TRUE-N (no newcomer)	2596.4	[2223.2, 2964.0]	50.7	0.805	0.663	0.398	4.2
Observation-Manipulated	Blind	-24965.6	[-25318.0, -24577.2]	196.3	0.335	0.517	0.310	7.3
Observation-Manipulated	TTB	-28713.6	[-29066.4, -28348.8]	222.3	0.229	0.858	0.515	999.0
Observation-Manipulated	MOO (Tchebycheff)	-710.4	[-1292.0, -134.8]	66.3	0.757	0.645	0.387	594.2
Utility-Trust-Misalignment	TRUE	2482.9	[2205.5, 2761.0]	55.2	0.750	0.619	0.380	0.2
Utility-Trust-Misalignment	TRUE-C (no constraints)	3506.0	[3192.1, 3835.0]	53.7	0.784	0.661	0.397	0.7
Utility-Trust-Misalignment	TRUE-E (no exploration)	2340.4	[2023.2, 2664.0]	55.9	0.749	0.620	0.381	0.1
Utility-Trust-Misalignment	TRUE-N (no newcomer)	3022.6	[2660.6, 3381.8]	55.3	0.783	0.676	0.406	339.7
Utility-Trust-Misalignment	Blind	-21783.3	[-22110.9, -21456.9]	206.3	0.339	0.535	0.321	6.2
Utility-Trust-Misalignment	TTB	167.4	[-540.1, 793.5]	68.0	0.727	0.722	0.434	780.2
Utility-Trust-Misalignment	MOO (Tchebycheff)	1250.3	[661.9, 1764.8]	63.3	0.747	0.699	0.419	772.9

Baseline — Time-Series Analysis

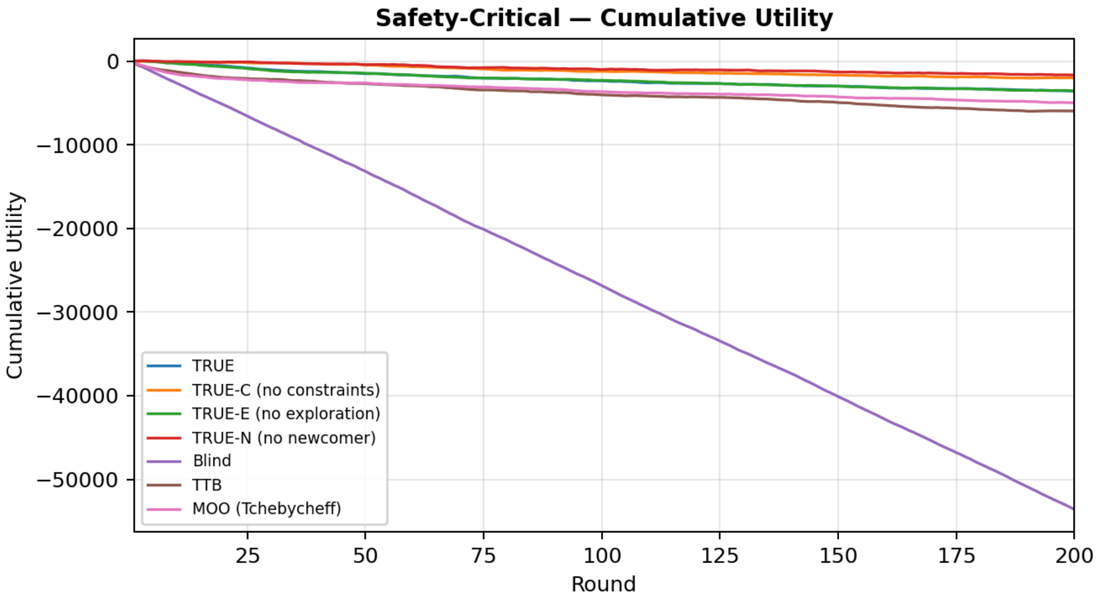


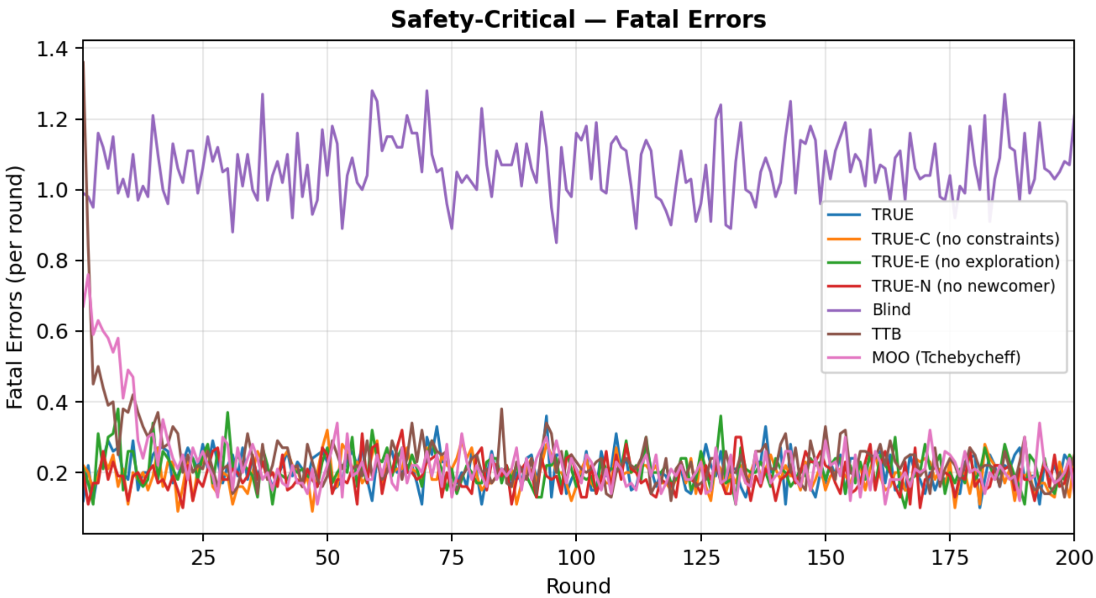


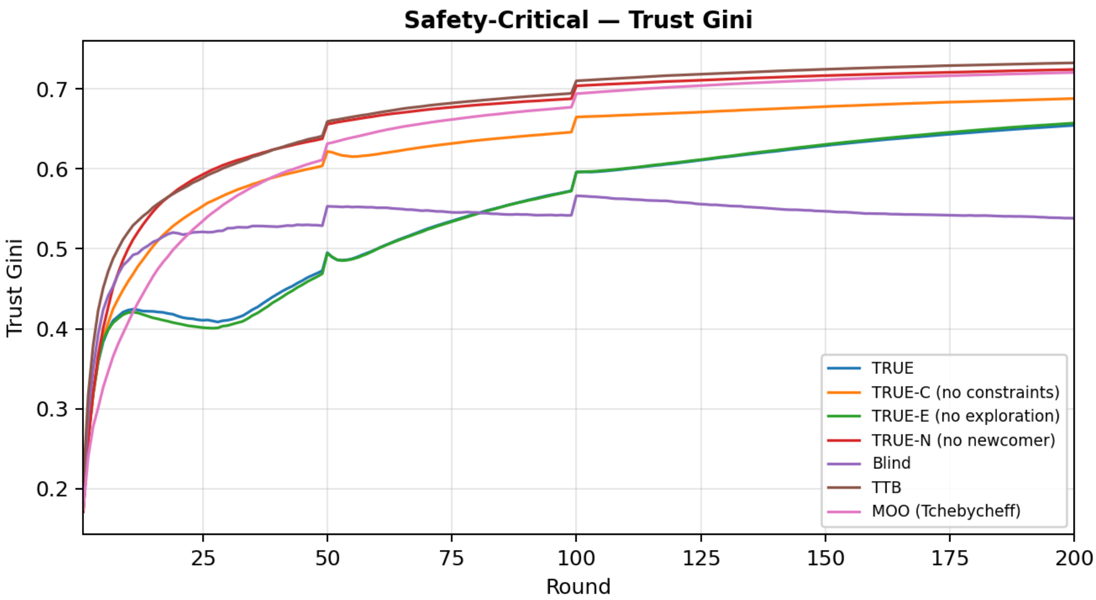

Baseline — Trust Gini
Trust Gini
Round
TRUE
TRUE-C (no constraints)
TRUE-E (no exploration)
TRUE-N (no newcomer)
Blind
TTB
MOO (Tchebycheff)

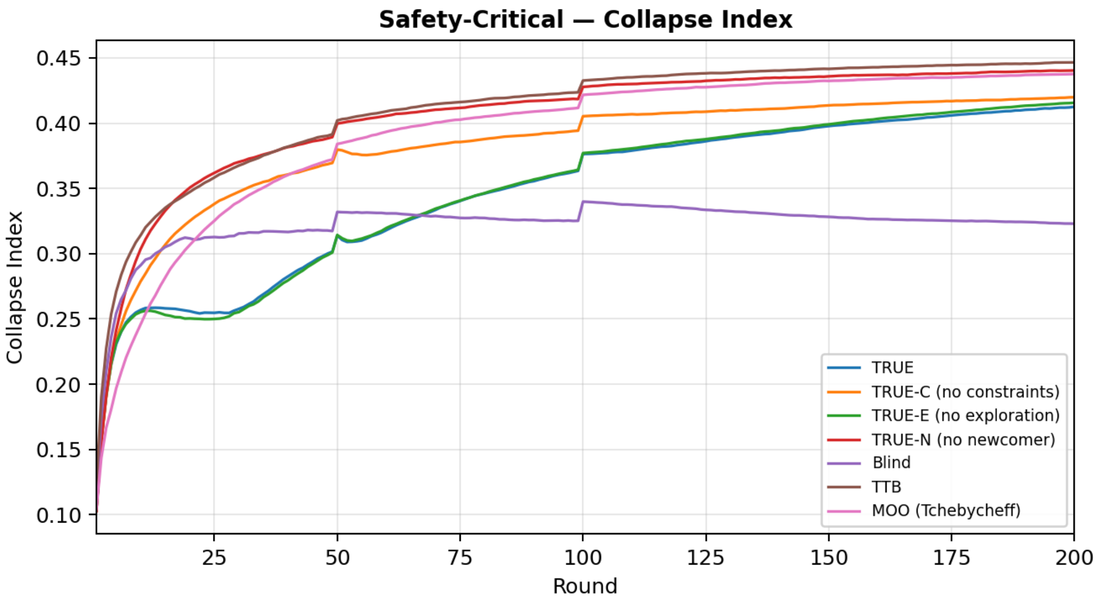

Baseline — Collapse Index
Collapse Index
Round
TRUE
TRUE-C (no constraints)
TRUE-E (no exploration)
TRUE-N (no newcomer)
Blind
TTB
MOO (Tchebycheff)

Safety-Critical — Time-Series Analysis

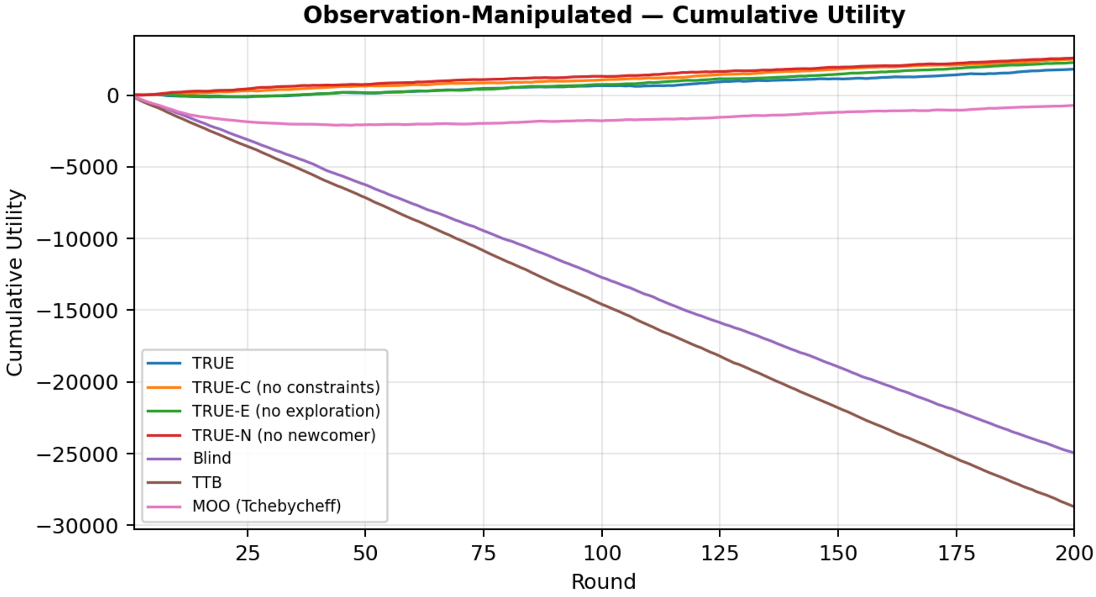


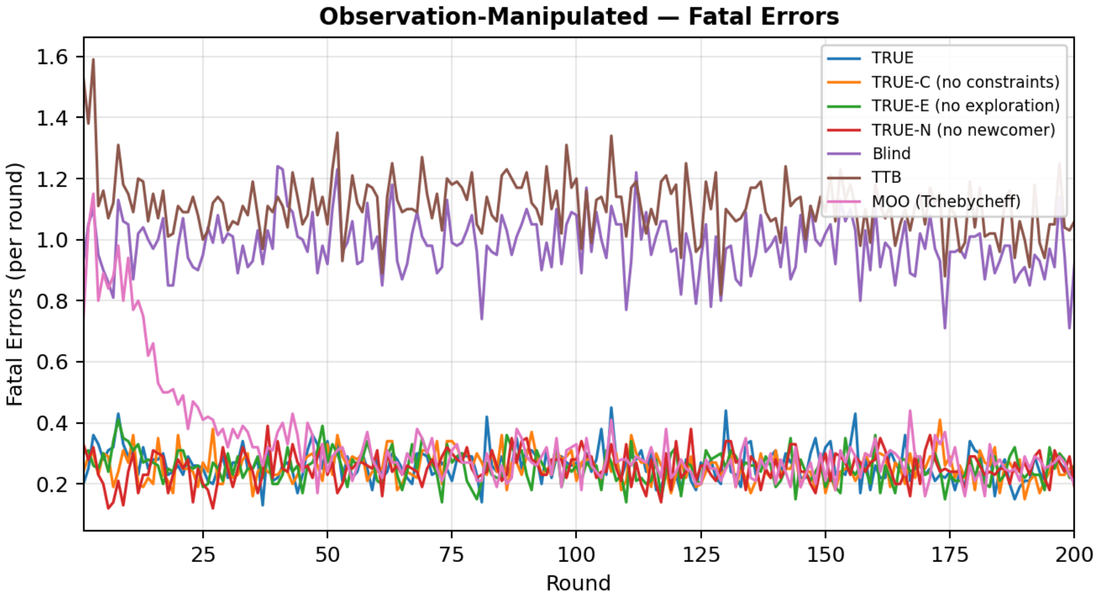


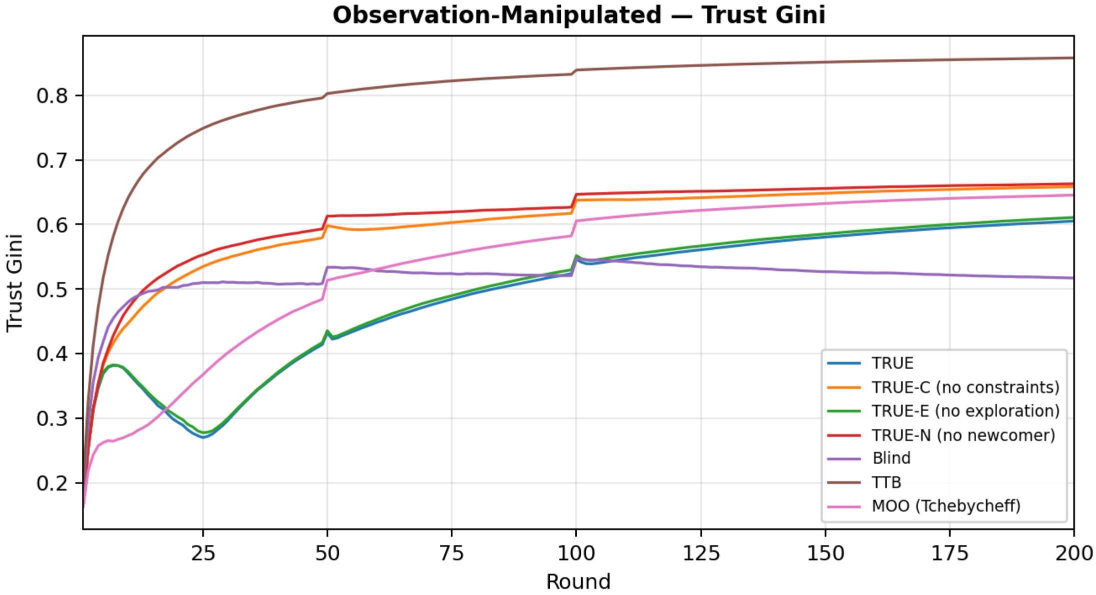

Safety-Critical — Trust Gini
Trust Gini
Round
TRUE
TRUE-C (no constraints)
TRUE-E (no exploration)
TRUE-N (no newcomer)
Blind
TTB
MOO (Tchebycheff)

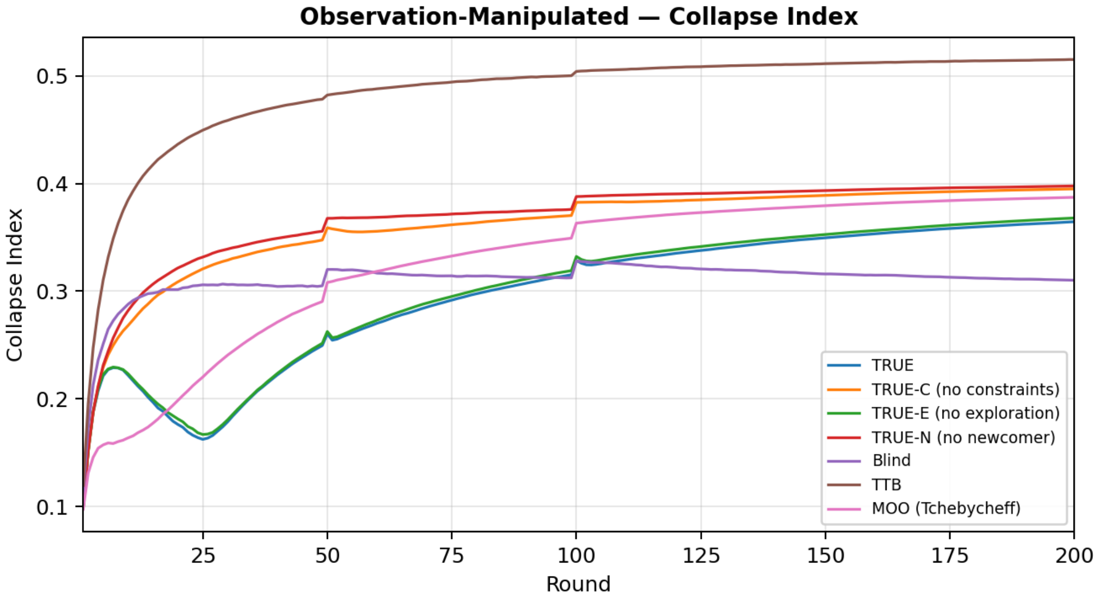

Safety-Critical — Collapse Index
Collapse Index
Round
TRUE
TRUE-C (no constraints)
TRUE-E (no exploration)
TRUE-N (no newcomer)
Blind
TTB
MOO (Tchebycheff)

Observation-Manipulated — Time-Series Analysis

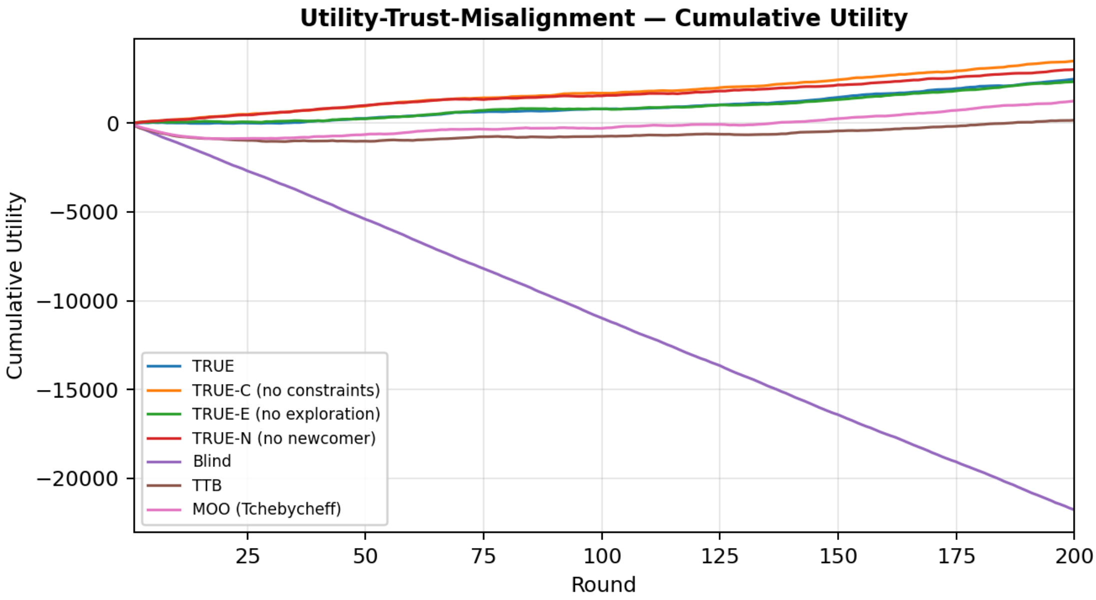


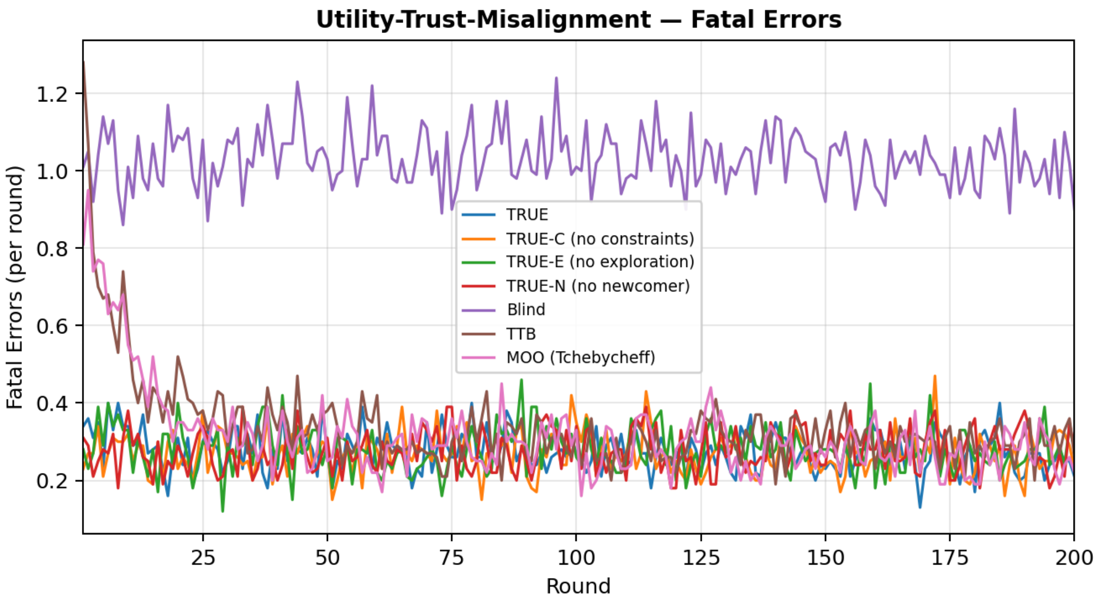


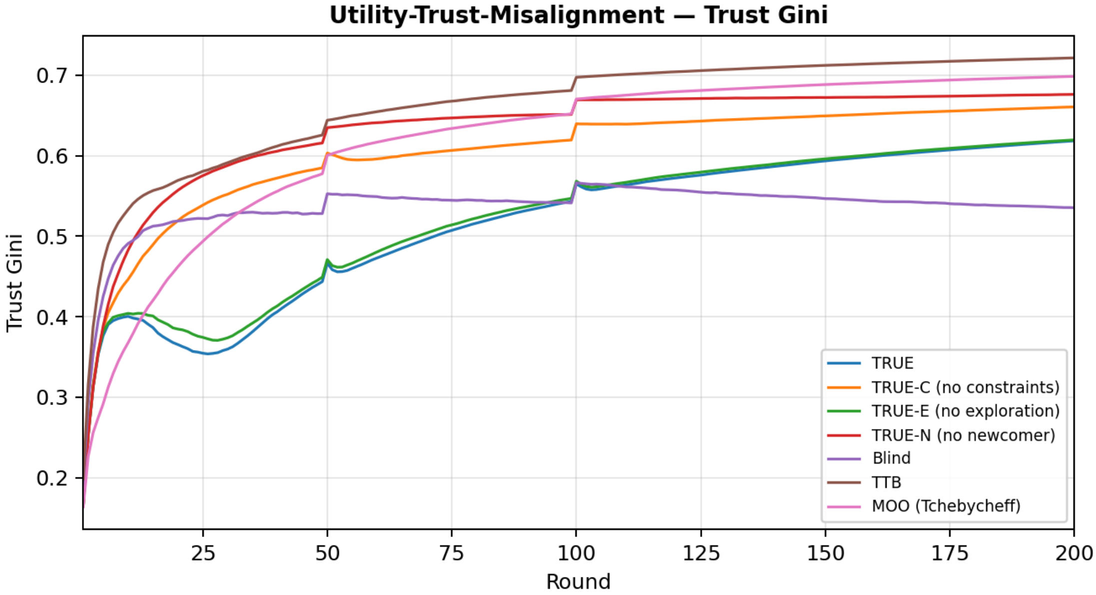

Observation-Manipulated — Trust Gini
Trust Gini
Round
TRUE
TRUE-C (no constraints)
TRUE-E (no exploration)
TRUE-N (no newcomer)
Blind
TTB
MOO (Tchebycheff)

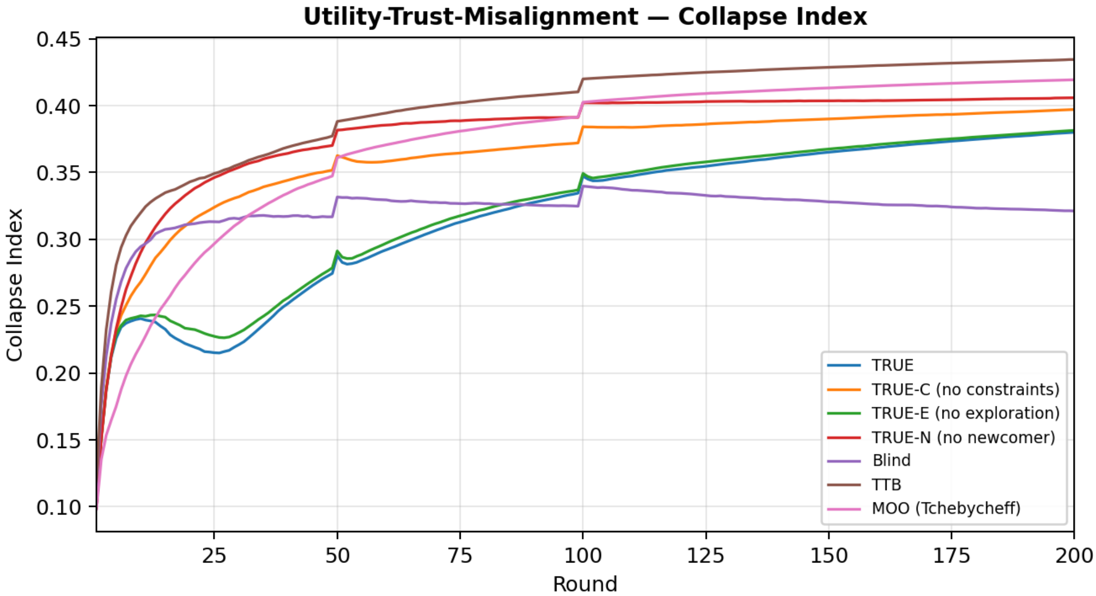

Observation-Manipulated — Collapse Index
Collapse Index
Round
TRUE
TRUE-C (no constraints)
TRUE-E (no exploration)
TRUE-N (no newcomer)
Blind
TTB
MOO (Tchebycheff)

Utility-Trust-Misalignment — Time-Series Analysis






Utility-Trust-Misalignment — Trust Gini
Trust Gini
Round
TRUE
TRUE-C (no constraints)
TRUE-E (no exploration)
TRUE-N (no newcomer)
Blind
TTB
MOO (Tchebycheff)


Utility-Trust-Misalignment — Collapse Index
Collapse Index
Round
TRUE
TRUE-C (no constraints)
TRUE-E (no exploration)
TRUE-N (no newcomer)
Blind
TTB
MOO (Tchebycheff)

Hypothesis Tests (Selected)

All p-values reported with Bonferroni correction across the full test battery.

Scenario	Hypothesis	Mean Diff	t	p_t(Bonf)	Cohen d
Baseline	TRUE_U > Blind_U	24829.8	123.10	0.0000	12.3
Baseline	TRUE_Fatal < Blind_Fatal	150.3	123.37	0.0000	12.3
Baseline	TRUE_U > TTB_U	2474.8	4.51	0.0008	0.5
Baseline	TRUE_Fatal < TTB_Fatal	12.7	4.51	0.0008	0.5
Baseline	TRUE_U > MOO_U	1880.8	4.97	0.0001	0.5
Baseline	TRUE_Fatal < MOO_Fatal	10.2	5.70	0.0000	0.6
Baseline	TRUE_Collapse < TTB_Collapse	0.0	19.83	0.0000	2.0
Baseline	TRUE_A9 < Blind_A9	6.8	10.05	0.0000	1.0
Observation-Manipulated	TRUE_U > Blind_U	26781.2	110.03	0.0000	11.0
Observation-Manipulated	TRUE_Fatal < Blind_Fatal	143.8	103.00	0.0000	10.3
Observation-Manipulated	TRUE_U > TTB_U	30529.2	134.71	0.0000	13.5
Observation-Manipulated	TRUE_Fatal < TTB_Fatal	169.8	93.74	0.0000	9.4
Observation-Manipulated	TRUE_U > MOO_U	2526.0	7.74	0.0000	0.8
Observation-Manipulated	TRUE_Fatal < MOO_Fatal	13.8	8.90	0.0000	0.9
Observation-Manipulated	TRUE_Collapse < TTB_Collapse	0.2	145.37	0.0000	14.5
Observation-Manipulated	TRUE_A9 < Blind_A9	7.3	9.31	0.0000	0.9
Safety-Critical	TRUE_U > Blind_U	49997.8	138.89	0.0000	13.9
Safety-Critical	TRUE_Fatal < Blind_Fatal	171.0	138.19	0.0000	13.8
Safety-Critical	TRUE_U > TTB_U	2383.2	3.36	0.1014	0.3
Safety-Critical	TRUE_Fatal < TTB_Fatal	6.7	3.72	0.0256	0.4
Safety-Critical	TRUE_U > MOO_U	1376.6	3.10	0.2483	0.3
Safety-Critical	TRUE_Fatal < MOO_Fatal	4.4	3.80	0.0183	0.4
Safety-Critical	TRUE_Collapse < TTB_Collapse	0.0	15.87	0.0000	1.6
Safety-Critical	TRUE_A9 < Blind_A9	7.5	7.80	0.0000	0.8
Utility-Trust-Misalignment	TRUE_U > Blind_U	24266.2	117.44	0.0000	11.7
Utility-Trust-Misalignment	TRUE_Fatal < Blind_Fatal	151.2	103.42	0.0000	10.3
Utility-Trust-Misalignment	TRUE_U > TTB_U	2315.5	5.89	0.0000	0.6
Utility-Trust-Misalignment	TRUE_Fatal < TTB_Fatal	12.8	6.58	0.0000	0.7
Utility-Trust-Misalignment	TRUE_U > MOO_U	1232.6	3.93	0.0107	0.4
Utility-Trust-Misalignment	TRUE_Fatal < MOO_Fatal	8.1	5.44	0.0000	0.5
Utility-Trust-Misalignment	TRUE_Collapse < TTB_Collapse	0.1	23.57	0.0000	2.4
Utility-Trust-Misalignment	TRUE_A9 < Blind_A9	6.0	9.33	0.0000	0.9

Interpretation & Limitations

Task-flow pairing: Within each run all groups see the same module sequence. Differences are therefore attributable to selection mechanisms, not task luck.

Observation uniformity: A8 surface-quality bonus is uniform across groups. The only A8 asymmetry in TTB is its scoring-function bonus (hard-coded trust-objective weight), which is a mechanism-level difference. MOO has no hard-coded A8 bonus; any A8 advantage emerges naturally from high surface-quality observations.

MOO vs TTB: The weighted Tchebycheff MOO baseline includes explicit diversity objective (selection concentration penalty) and per-candidate-pool normalization. It consistently outperforms TTB but still falls short of TRUE, confirming that the problem is "trust-as-objective" rather than the specific aggregation function.

Ablation surprise: TRUE-C and TRUE-N outperform full TRUE in cumulative utility under current parameters, suggesting that constraint filtering and newcomer protection may be overly conservative. However, TRUE-N shows dramatically higher A9 cold-start delay, confirming that newcomer protection serves its intended purpose.

Bonferroni correction: Conservative; hypotheses remaining significant after correction are robust.